

# FASTrack: A co-designed cloud-based planning support system for interactive cycling infrastructure scenario assessment

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## Abstract

Testing transport interventions using city-scale transport models is typically resource intensive, requiring specialist expertise, substantial data and computational capacity. As a result, such modelling is often concentrated on major projects and larger metropolitan areas, while smaller interventions and mid-sized regional cities have limited access to comparable modelling capabilities. This is particularly relevant to active travel, where cycling infrastructure decisions frequently need to be made with relatively limited modelling and decision-support capacity. This study presents FASTrack, a cloud-based planning support software system that integrates agent-based modelling, behavioural simulation, and interactive scenario visualisation to enable proposed cycling infrastructure assessment. A case study is applied to the City of Greater Bendigo, a mid-sized regional Australian city representative of the planning contexts that existing platforms often overlook. FASTrack was developed through a co-design process over three years with local government planning partners, to address the need to rapidly assess infrastructure scenarios, quantify changes in cycling demand and behaviour, and inform infrastructure planning and investment appraisal. These requirements underpin its design, including the use of probabilistic modelling and a cloud-based workflow designed to support rapid, on-demand scenario testing with minimal specialist modelling expertise. Scenario outcomes are presented through interactive dashboard-style interfaces that detail estimated changes in network use, level of traffic stress, mode shift, cycling usage, emissions, and cost-benefit analysis aligned with national appraisal guidelines. This study demonstrates how co-design can directly shape the development of planning support systems by translating practitioner requirements into accessible modelling platforms for cycling infrastructure planning.

**Keywords:** active transport; bicycle infrastructure investment; user-interaction; planning processes; decision-support framework

## 1 Introduction

Active travel, referring to forms of transportation that rely on human-powered mobility such as walking and cycling, provides many health and environmental benefits compared with vehicular mobility (Ding et al., 2024; Nieuwenhuijsen, 2021). Cycling in particular has attracted considerable policy attention and infrastructure investment across Europe, North America, Australasia, and East Asia as governments seek to reduce private vehicle dependence and increase physical activity at the population level (Harris and Roberts, 2026; Pucher and Buehler, 2017). However, active travel investment has been predominantly focused on large metropolitan areas, with comparatively limited attention given to regional cities (Asker et al., 2025; Bosworth et al., 2020). Compared to mono-centric metros on one hand and dispersed rural areas on the

other, regional cities strike a fine balance for active travel potential due to their smaller scales and shorter daily travel distances between key destinations respectively. Yet, this potential remains largely under-explored, with regional cities lacking safe cycling infrastructure (King et al., 2020) and receiving lower investment relative to metro areas. This metropolitan focus is also evident in the modelling tools, planning support systems, and decision-making frameworks developed to support cycling investment. These have predominantly been designed for large metropolitan regions, where data availability, institutional capacity, and research attention are concentrated (Ito et al., 2023; Lowry et al., 2016; Beecham and Wood, 2014). Consequently, regional contexts remain comparatively underrepresented in both active travel research and the development of decision-support systems (Bastariento et al., 2023). This creates a practical gap for regional governments, which may have limited access to planning support systems that account for the characteristics and resource constraints of regional cities. Without appropriate analytical platforms, regional governments may face greater difficulty evaluating, prioritising, and building evidence-based cases for cycling infrastructure investment, limiting their capacity to realise the potential for active travel in these communities.

Modelling of cycling behaviour has become increasingly sophisticated, moving beyond the estimation of existing cycling demand towards the simulation of behavioural responses to infrastructure and policy interventions. One prominent example is the Propensity to Cycle Tool (PCT), an open-source planning support system developed for England and Wales that uses micro-simulation to estimate cycling uptake under alternative scenarios and assess associated health, environmental, and distributional impacts (Lovelace et al., 2017; Woodcock et al., 2021), which has since been adapted for other European contexts (Félix et al., 2025; Lovelace et al., 2024). Other recent research has placed greater emphasis on representing the behavioural and network characteristics that influence cyclists’ route choices. Studies have demonstrated the value of incorporating detailed infrastructure and environmental attributes, including facility type, traffic conditions, gradient, land use, and route surroundings (Jafari et al., 2025a; Gao et al., 2018), as well as GPS-based route-choice models (Meister et al., 2024; Łukawska et al., 2023). Other approaches have integrated cycling route choice directly into travel demand models (Tarkkala et al., 2024), while agent-based approaches have been used to simulate the behavioural and network effects of infrastructure interventions (Jafari et al., 2025b). Collectively, these developments demonstrate a shift towards increasingly detailed representations of cycling behaviour and infrastructure.

Improvements in modelling have also facilitated the development of platforms to inform infrastructure planning and scenario assessment. For example, demand-driven approaches have been developed to design bicycle networks based on spatial distributions of cycling demand and estimated route-choice behaviour (Steinacker et al., 2022). Recent Australian work has combined behavioural modelling, synthetic populations, and spatial data within an interactive tool that allows users to test hypothetical infrastructure changes and estimate their effects on cycling uptake (Roberts et al., 2026). As cycling modelling becomes increasingly complex, the challenge therefore becomes the translation of model outputs into usable information that addresses the practical needs of their intended users. This highlights the importance of working with end users through the design and development processes, particularly where models are intended to support significant infrastructure decisions.

Co-design is an approach to system and service development in which intended users and other relevant stakeholders actively define problems, identify requirements, generate solutions, and evaluate outcomes throughout the development process. In the context of planning support and scenario-testing platforms, co-design is particularly valuable because these systems translate complex analytical models into interfaces and outputs intended to support real-world decision-making. Their effectiveness therefore depends not only on the technical capabilities of the underlying models, but also on whether the scenarios, functionality, outputs, and interactions reflect the needs and decision-making processes of their intended users (Rittenbruch et

al., 2022). By involving stakeholders throughout development, co-design provides a mechanism for combining technical and domain expertise with local knowledge and practical planning requirements, while allowing system concepts and functionality to be iteratively tested and refined.

The application of co-design to planning and environmental decision-support systems has demonstrated its potential to improve the relevance, usability, transparency, and adoption of digital platforms. Previous studies have applied co-design approaches to urban planning support systems (Flacke et al., 2025; Dane et al., 2024; Rittenbruch et al., 2022; Lock et al., 2021), land value analysis and scenario development (Pettit et al., 2020), information model-based land use planning (Nummi and Hapuoja, 2025), urban heat and climate services (Ramsey et al., 2024), stakeholder engagement platforms for environmental planning (Aguilar et al., 2023), and spatially-enabled climate-security risk assessment (Young et al., 2023). Collectively, these studies demonstrate that co-design can extend beyond interface development by influencing the underlying objectives, functionality, data requirements, and outputs of planning support systems. This is particularly important for scenario-based platforms, where stakeholders must translate complex model outputs into information that is meaningful and actionable within their existing planning and decision-making processes.

Successful co-design requires sustained and meaningful stakeholder participation. Foundational activities include identifying and recruiting stakeholders with relevant knowledge, responsibilities, and perspectives (Wacnik et al., 2025), followed by opportunities for participants to define problems, generate ideas, evaluate alternatives, and provide feedback throughout development (Blomkamp, 2018). Iterative engagement and the use of multiple participation techniques can further refine system requirements as stakeholder understanding and priorities develop (Wacnik et al., 2025). Such an approach is particularly relevant when developing planning support systems that incorporate complex analytical models, as it provides a means of ensuring that methodological capabilities are translated into functionality and outputs appropriate for practical planning contexts.

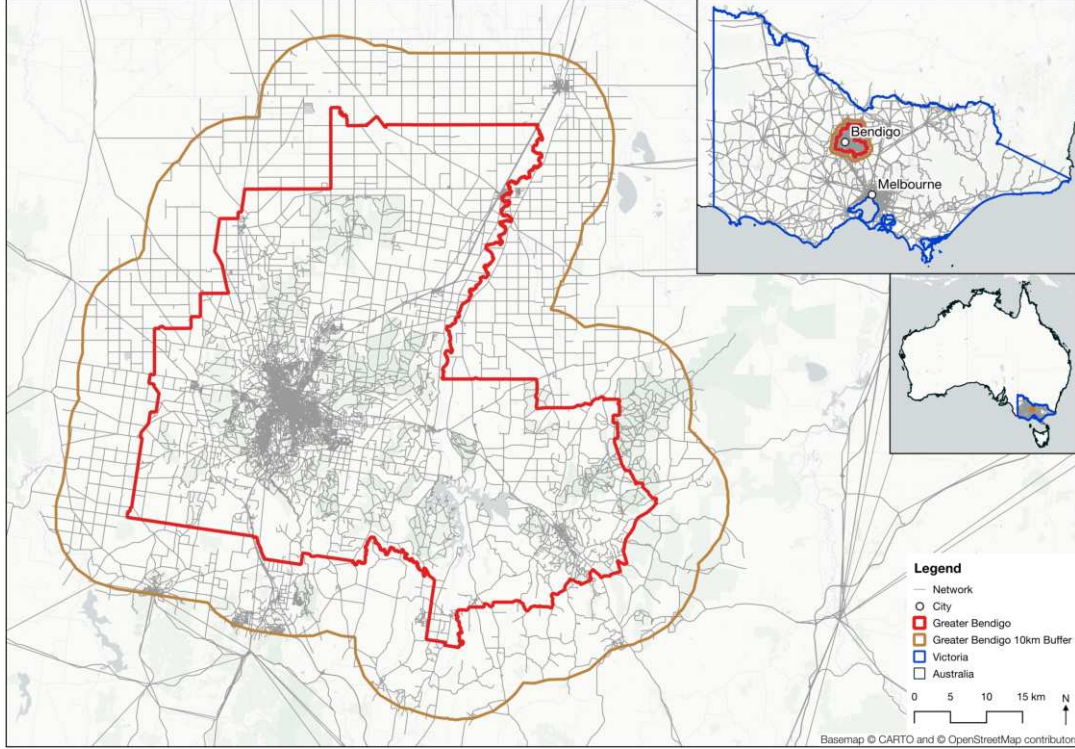
In Australia, the Victorian State Government has adopted a formal commitment to increase walking and cycling trips from 18% in 2021 to 25% by 2030 (Victorian State Government Department of Energy, Environment and Climate Action, 2025). Increasing cycling infrastructure investments in regional areas is of particularly high strategic importance to promote active travel to a broader range of users. To support this initiative, a partnership of researchers and local and state government representatives was formed to develop a planning support system to enable rapid assessment of proposed cycling infrastructure uptake and related cost-benefit analysis.

The aim of this study is therefore to demonstrate how a co-design approach can be used for translating practitioner requirements into design choices of planning support systems. FAS-Track provides the empirical case through which this approach is examined, with focus on cycling infrastructure planning in a regional city. Specifically, this study seeks to demonstrate how sustained engagement with planning practitioners can reveal and consolidate practical decision-support requirements, examine how these practitioner requirements can be translated into methodological, computational, interface, and decision-support design choices within a complex planning support system, and derive broader lessons and transferable principles for the co-design and development of practitioner-facing planning support systems.

## 2 Methodology

### 2.1 Regional Context

The City of Greater Bendigo was selected as a pilot case study because of its practical relevance as a major regional city requiring planning approaches that differ from those used in metropoli-



**Figure 1:** Greater Bendigo study area and 10 km buffer, geographical proximity to Melbourne, and context within the state of Victoria and Australia.

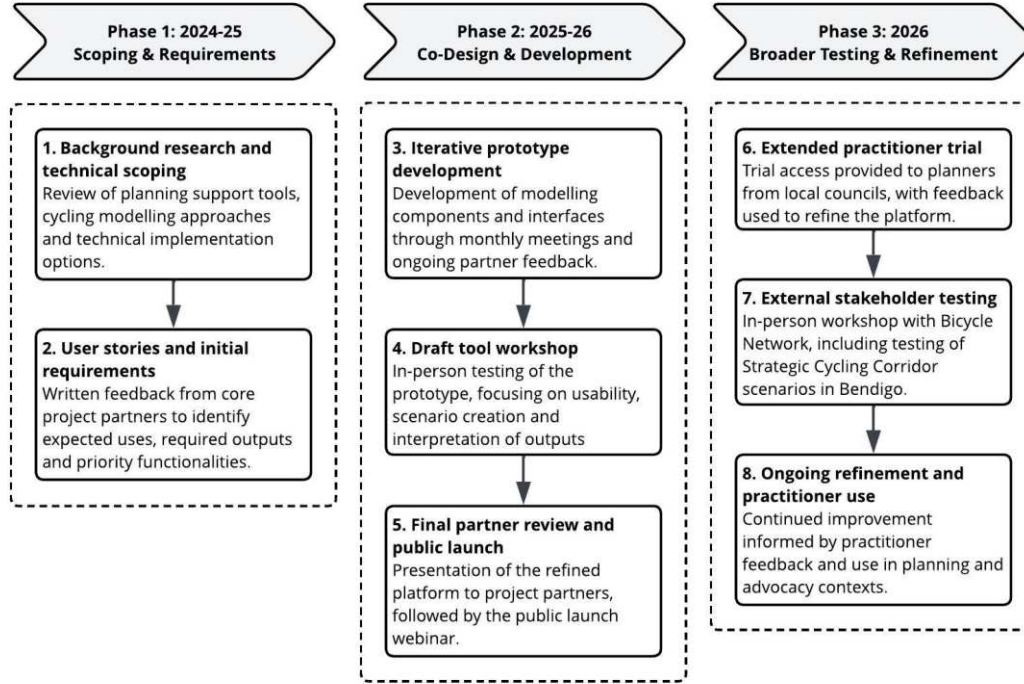
tan contexts. Located approximately 150 km north-west of Melbourne, Greater Bendigo is a major regional centre and service hub for north-central Victoria. With a bicycle mode share of approximately 1.26% for commuting and journey-to-work trips (Tiwari et al., 2026a), there is significant potential to capture latent demand for cycling.

The municipality covers approximately 3,000 km<sup>2</sup> and had a population of 121,470 at the 2021 Australian Census (Australian Bureau of Statistics, 2021). The municipality encompasses the urban centre of Bendigo as well as a number of smaller surrounding communities, creating a diverse spatial context in which transport connections between residential areas, employment, education, retail, and other services are all important to local mobility. As shown in Figure 1, a 10 km buffer of Greater Bendigo was used to retain network detail in the periphery of the study area and reduce edge effects.

## 2.2 Co-design and Iterative Development

The development of FASTrack followed a co-design approach in which project partners were engaged throughout a three-year timeline to identify requirements, evaluate emerging functionality, and iteratively refine the system. The process brought together the technical expertise of the research team with the practical knowledge and planning needs of government representatives. Rather than treating stakeholder engagement as a discrete consultation exercise, feedback was incorporated throughout development, with successive versions of FASTrack used to identify and address emerging requirements. Key stages of the co-design and development process are presented chronologically in Figure 2.

Following the identification and confirmation of key project partners, the initial stage of co-design focused on establishing the functional and analytical requirements of FASTrack. The researchers first reviewed existing interactive platforms for active travel modelling and investigated potential approaches for the technical deployment of FASTrack. A user stories approach was subsequently adopted to translate stakeholder needs into specific system requirements



**Figure 2:** FASTrack co-design and development process, progressing from initial scoping and requirements elicitation through iterative co-design to broader practitioner testing and ongoing refinement.

(Aguilar et al., 2020; Brhel et al., 2015; Cohn, 2004). At the end of Phase 1, project partners completed a structured written feedback form, which asked participants to identify their expectations for FASTrack, the types of cycling infrastructure scenarios they needed to assess, and the outputs and indicators required to support planning and investment decisions. The resulting user stories established an initial set of functional and analytical requirements that guided the development of the first working version of the platform.

Development then proceeded iteratively, with partners engaged through regular feedback cycles throughout the remainder of the project. Monthly project meetings provided a consistent mechanism for stakeholder participation, during which the research team presented updates related to emerging functionality, modelling outputs, and interface prototypes for discussion and review. Feedback from these sessions was used to refine existing functionality and identify additional requirements as the capabilities of FASTrack developed. Midway through Phase 2, an in-person workshop provided an opportunity for project partners to interact with and trial an initial draft version of FASTrack. The workshop focused on testing the end-to-end workflow from an end user’s perspective, including the creation and modification of cycling infrastructure scenarios, interpretation of model outputs, and use of the resulting information for planning purposes. Feedback from this workshop informed further refinement of both the interface and analytical outputs.

Following these iterative development activities, a final partner review was undertaken at the end of Phase 2. Project partners were provided with the near-final version of FASTrack to review the functionality, outputs, and overall suitability for practical planning applications. Final feedback was incorporated into the released version of the platform, which was formally introduced through a public launch webinar. This stage marked the transition from development with the core project partners to broader use of FASTrack as a planning support resource.

In Phase 3, FASTrack underwent an extended trial involving planners from local councils across Greater Melbourne. This exercise was intended to examine the transferability and practical usability of the platform beyond the core project partners and to identify requirements

that may not have emerged during development within a single local government context. Participating planners were provided with access to FASTrack and invited to use the platform to develop and assess cycling infrastructure scenarios in Greater Bendigo. Feedback from this extended trial was used to identify opportunities to further improve usability, interpretation of outputs, workflow, and applicability of the scenario generation and reporting functions. The extended trial therefore provided an additional practitioner evaluation and refinement stage in which FASTrack was assessed by practitioners who had not been directly involved in its initial development.

External stakeholder testing was subsequently undertaken through a workshop with Bicycle Network, an independent Australian cycling advocacy organisation (Bicycle Network, 2026). This workshop provided a perspective distinct from that of local and state government planners and was intended to assess whether the outputs of FASTrack could generate tangible evidence to support various cycling advocacy initiatives. The workshop included discussion and testing of the use of FASTrack for assessing proposed and existing separated cycling corridors in Greater Bendigo. Feedback from this session was used to identify further opportunities for refinement and to assess the broader applicability of FASTrack beyond the original pilot context.

Collectively, the successive stages of partner engagement, iterative development, extended practitioner trial, and external stakeholder testing provided a progressive process for refining FASTrack from an initial set of user requirements into a planning support system capable of supporting cycling infrastructure scenario assessment across different planning contexts.

## 2.3 Identification of Practitioner Priorities

Practitioner priorities were derived by synthesising requirements and feedback collected throughout the co-design process, including the initial user stories and written feedback form, regular project meetings, prototype testing, the Phase 2 workshop, final partner review, the extended practitioner trial, and external stakeholder testing (see Supplementary material). Feedback was reviewed across these activities to identify requirements that recurred across engagement stages or were considered important to the practical application of FASTrack.

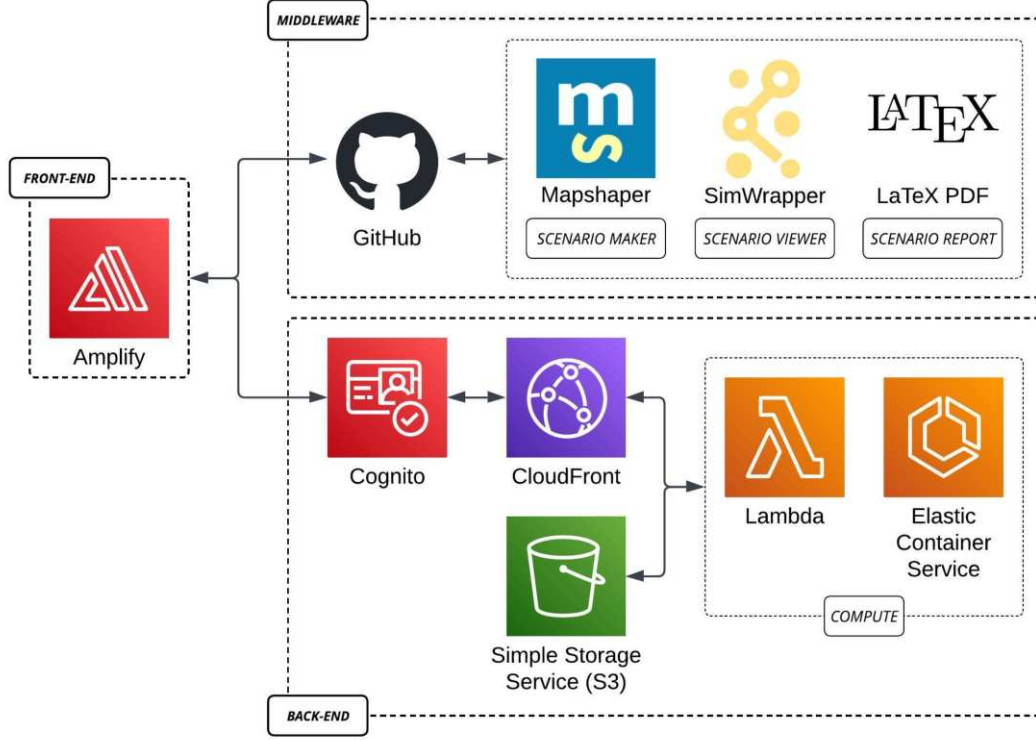
Requirements were grouped according to the underlying planning need they addressed rather than the specific software feature through which they were implemented. These resulting priorities were then used to examine how practitioner requirements were translated into the methodological, technical, interface, and decision-support components of FASTrack.

## 2.4 FASTrack System Design and Cloud-based Implementation

The following sections describe how the practitioner priorities identified through co-design were translated into methodological, technical, interface, and decision-support components.

FASTrack was designed as a cloud-based planning support system that connects an interactive scenario development interface with automated modelling, visualisation, and reporting workflows. The system architecture was designed to allow users to create and modify cycling infrastructure scenarios without requiring local installation of modelling software or specialist computational resources. Amazon Web Services (AWS) was selected as the cloud technology provider because of its scalable computing, storage, authentication, and hosting capabilities. Figure 3 presents the technical architecture, showing the relationships between the user interface, cloud-based processing, modelling components, and reporting outputs.

The workflow begins with AWS Cognito managing user registration and authentication, while Amazon Simple Storage Service (S3) provides private storage for scenario data and user-generated files. The browser-based Scenario Maker provides an interactive environment in which users can explore the baseline cycling network and draw or edit proposed infrastructure. The open-source Mapshaper package (Bloch, 2025) was modified and integrated into the Scenario Maker to support spatial data manipulation.



**Figure 3:** Technical architecture of FASTrack, showing the relationships between the back-end, middleware, and front-end components.

Once a scenario is submitted by the user, the modified network is passed to an automated sequence of cloud-based processing functions. AWS Lambda and Amazon Elastic Container Service (ECS) provide the computational infrastructure for executing the modelling and analysis components.

The workflow converts the user-edited network into MATSim-compatible inputs, generates scenario catchments, prepares travel plans, applies the mode-choice and cycling route-choice models, and produces network, spatial, emissions, and cost-benefit outputs. The cloud-based workflow allows computationally intensive processes to be executed remotely, so users do not need to manage the underlying modelling environment or execute individual modelling stages manually. Each compute module is detailed in Table 1.

The resulting outputs are processed into interactive dashboards through the Scenario Viewer. SimWrapper (Charlton, 2025; Charlton and Sana, 2023), an open-source visualisation framework for MATSim modelling, was integrated into FASTrack to provide interactive maps, charts, and summary indicators. The dashboards organise outputs into categories relevant to cycling infrastructure assessment, including network infrastructure, mode shift, cycling exposure, emissions, and cost-benefit analysis. Following methods by Jafari and Pemberton (2024), LTS changes are also included to quantify changes in cycling safety based on the network attributes of cycling infrastructure type, road type, traffic volume, and speed limit.

The practitioner requirement for business-case evidence led to the inclusion of a cost-benefit analysis (CBA) module. Estimated infrastructure costs are combined with modelled changes in cycling activity and associated benefits are calculated to generate scenario-level economic indicators aligned with Australian national appraisal guidance. Integrating CBA with the behavioural and network modelling workflow allows infrastructure scenarios to be assessed for both their transport effects and potential economic returns.

The need for clear and communicable outputs as identified by practitioners led to the development of a scenario reporting system. Using the LaTeX typesetting system, FASTrack

**Table 1:** Cloud-based compute modules and functions used to process network edits and generate a FASTrack scenario.

| Compute module               | Purpose                                                                                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cognito Authorisation        | Manages user pool sign-up, log-in/out, and access tokens                                                                                                                                             |
| Network SHP to XML           | Converts user-created scenario network edits from <code>.shp</code> to <code>.xml</code> format and calculates change in Level of Traffic Stress (LTS) based on network attribute edits <sup>1</sup> |
| Dashboard Generator          | Creates the Scenario Viewer dashboards based on a pre-configured template                                                                                                                            |
| Scenario Catchment           | Buffers edited network links to create 3 km and 1 km scenario catchment regions                                                                                                                      |
| Plan XML                     | Creates a MATSim-compatible <code>.xml</code> plan file based on scenario network edits                                                                                                              |
| Mode Shift                   | Generates scenario mode share and mode shift statistics relative to the baseline                                                                                                                     |
| Link Stats                   | Generates network link-level mean daily cycling volume and cycling travel-time changes                                                                                                               |
| SA1 Aggregation <sup>2</sup> | Spatially aggregates link-level changes in mean daily cycling volume and travel time to SA1 level                                                                                                    |
| CBA Trip Changes             | Computes changes in cycling and car trips between the baseline and scenario in terms of total number, travel time, and distance for cost-benefit analysis                                            |
| Emissions                    | Generates emissions outputs for four common pollutant types                                                                                                                                          |

<sup>1</sup> LTS determined by methods outlined in (Jafari and Pemberton, 2024).

<sup>2</sup> Statistical Area Level 1 (SA1) units defined by the Australian Bureau of Statistics typically represent 200–800 residents, with an average of around 400 residents.

generates a consolidated PDF report that combines key outputs from the modelling, dashboards, and cost-benefit analysis into a format that can be retained, shared, and incorporated into planning and business-case processes. The complete workflow therefore connects scenario creation, cloud-based modelling, spatial and behavioural analysis, economic appraisal, visualisation, and reporting within a single browser-based planning support system.

## 2.5 Probabilistic Scenario Assessment Framework

The practitioner requirement for rapid scenario assessment was particularly consequential for the modelling architecture. FASTrack builds on three previously developed complementary modelling components: 1) a regional synthetic population representing daily activity and travel patterns of agents (Tiwari et al., 2026a); 2) cluster-specific mode-choice models capturing differences in travel behaviour across population groups, incorporating both conventional travel determinants and cycling infrastructure characteristics (Tiwari et al., 2026c; Jafari et al., 2025a); and 3) a behaviourally informed cycling route-choice model reflecting preferences for route and infrastructure characteristics (Tiwari et al., 2026b). Together, these components form the Bendigo Agent-Based Model (B-ABM), which was developed using the Victorian Integrated Transport Model (VITM) 2018 synthetic population (Infrastructure Victoria, 2021), Victorian Integrated Survey of Travel and Activity (VISTA)<sup>1</sup> 2012-2023 data (Victorian State Government Department of Transport and Planning, 2024), and a cycling route choice public participatory map-based survey (Tiwari et al., 2025). For integration within FASTrack, the B-ABM was adapted for a cloud-based computing environment and its behavioural components were used within a probabilistic scenario-assessment workflow to enable rapid evaluation of user-defined cycling infrastructure scenarios.

The cloud-adapted B-ABM workflow begins by combining the user-edited network with the existing activity-based travel demand developed for Greater Bendigo (Tiwari et al., 2026a). Individual travel plans are generated for a 25% sample of the Greater Bendigo population. Each plan represents a daily sequence of activities and travel that includes trip purposes, origins and destinations, travel characteristics, and relevant demographic and household attributes. Trip purposes include home, work, study, shopping, pickup/drop-off/delivery, and social/recreational activities.

To account for behavioural heterogeneity, individuals are assigned to one of six demographic and behavioural clusters: male workers, female workers, male students, female students, other males, and other females. A separately estimated mode-choice model is then applied to each cluster (Tiwari et al., 2026a). For each scenario, travel attributes affected by the network intervention are recalculated and entered into the corresponding cluster-specific model to estimate the probabilities of travelling by car, bicycle, walking, or public transport. Because cycling infrastructure characteristics are included alongside conventional travel determinants, changes to the cycling network can directly influence the estimated probability of cycling.

The resulting cycling trips are subsequently assigned to the modified network using the behaviourally informed cycling route-choice model (Tiwari et al., 2026b), which accounts for preferences relating to route and infrastructure characteristics. The resulting scenario probabilities and route assignments are then compared with the corresponding B-ABM baseline to estimate changes in cycling behaviour and network outcomes associated with the proposed infrastructure intervention. This probabilistic approach avoids the need to repeatedly execute the full iterative B-ABM for each user-defined scenario, thereby reducing computational requirements while retaining the behavioural relationships represented in the underlying models.

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<sup>1</sup>VISTA is a household travel survey in Victoria in which all members of surveyed households report their travel over a single specified day. <https://public.tableau.com/app/profile/vista/vizzes/#!/>

## 2.6 Cost-Benefit Analysis and Land-Use Benefits

CBA is then performed to contextualise the monetary viability of scenario infrastructure and support business cases with tangible evidence. Results are framed in alignment with the Australian Transport Assessment and Planning (ATAP) guidelines Part M4 Active Travel (Australian Transport Assessment and Planning (ATAP) Steering Committee, 2023) to report the costs or benefits related to existing trips, new trips, unperceived vehicle operating costs and de-congestion, along with environmental and health system impacts. These are fed into standard CBA reporting statistics of net present value (NPV) and benefit-cost ratio (BCR).

An additional experimental land-use sensitivity component was developed to allow users to explore potential second-round transport benefits associated with large, city-shaping interventions. Consistent with ATAP O8 (Australian Transport Assessment and Planning (ATAP) Steering Committee, 2022), these outputs are intended for sensitivity analysis rather than as a substitute for dedicated Land Use and Transport Interaction (LUTI) modelling.

## 3 Results

### 3.1 Practitioner Priorities

The co-design process identified seven recurring priorities that shaped the development of FASTrack: rapid scenario testing, accessible scenario creation, representation of different cycling infrastructure types, understanding of spatial impacts, business-case evidence, clear and communicable outputs, and ease of access. These priorities extended beyond interface functionality to encompass the speed and accessibility of the modelling workflow, the representation and spatial distribution of impacts, and the translation of modelled outcomes into evidence relevant to infrastructure investment decisions. Table 2 summarises the priorities and their corresponding design responses and implementations. The priorities were translated into an interconnected set of methodological, technical, interface, and decision-support components rather than implemented as isolated software features.

#### 3.1.1 Rapid and Accessible Scenario Assessment

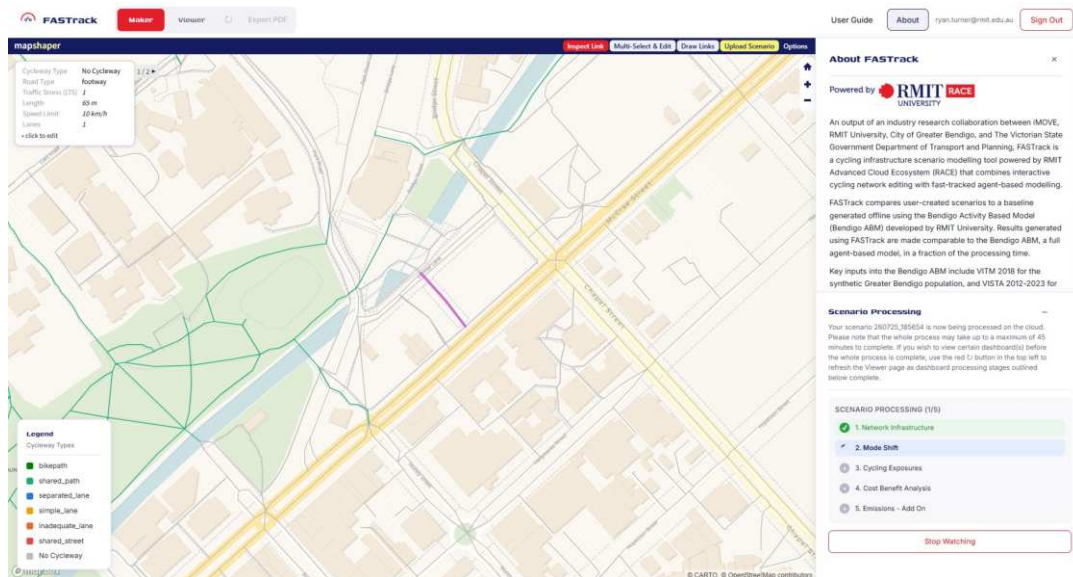
The practitioner requirement for rapid scenario testing led to changes in both the modelling approach and computational architecture of FASTrack. Rather than repeatedly executing the full iterative B-ABM for each user-defined scenario, FASTrack uses probabilistic mode-choice and route-choice estimates to provide a computationally efficient approximation of behavioural responses to network changes. This approach was integrated with an automated cloud-based processing workflow, allowing scenarios to be submitted and processed automatically.

The requirement for accessible scenario creation was addressed through the browser-based Scenario Maker (Figure 4). Users can modify the existing cycling network, remove links, and draw new infrastructure links while assigning relevant cycling infrastructure typologies and attributes. This allows infrastructure alternatives to be represented within the modelling framework without requiring users to directly manipulate network files or operate specialist software.

The resulting workflow connects scenario creation directly to model execution. Once a scenario is submitted, the modified network is automatically processed through the FASTrack modelling functions and the resulting outputs are made available through the Scenario Viewer. Typical scenario processing requires approximately 25–40 minutes, providing a substantially more rapid workflow than repeatedly executing the full baseline agent-based modelling process.

**Table 2:** Key priorities identified through the co-design process and their translation into FASTrack design responses and implemented features.

| Co-design priority                                       | Planning need                                                                                                     | Design response                                                                          | FASTrack implementation                                                                                                        |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Rapid scenario testing                                   | Explore and compare infrastructure options without repeatedly running a full agent-based model                    | Reduce computation time and automate the modelling workflow                              | Probabilistic mode-choice and route-choice modelling integrated with automated cloud processing                                |
| Accessible scenario creation                             | Create and modify cycling infrastructure scenarios without specialist GIS or transport modelling software         | Provide an intuitive map-based network editing environment                               | Scenario Maker supporting link drawing, network editing and modification of cycling infrastructure attributes                  |
| Representation of different cycling infrastructure types | Assess alternative infrastructure designs and their different cycling conditions                                  | Retain detailed cycling infrastructure attributes within scenario creation and modelling | Support for multiple cycleway typologies and associated Level of Traffic Stress attributes                                     |
| Understanding spatial impacts                            | Understand where the effects of infrastructure interventions occur rather than relying only on city-wide outcomes | Provide spatially disaggregated outputs at multiple scales                               | Greater Bendigo-wide outputs, 3 km and 1 km scenario catchments, network-link outputs, and SA1-level aggregation               |
| Business-case evidence                                   | Translate modelled impacts into indicators relevant to funding and investment decisions                           | Integrate economic appraisal within the scenario assessment workflow                     | ATAP-aligned cost-benefit analysis including NPV, BCR, health, decongestion, environmental and vehicle operating cost benefits |
| Clear and communicable outputs                           | Enable non-specialist users and decision-makers to interpret, communicate and share model results                 | Translate technical outputs into accessible visual summaries and reporting products      | Interactive dashboards together with automated generation of a professionally formatted downloadable PDF scenario report       |
| Ease of access                                           | Reduce reliance on specialist software, computing hardware and modelling expertise                                | Deliver the modelling workflow through an online environment                             | Browser-based platform using AWS for authentication, storage, automated computation and scenario delivery                      |



**Figure 4:** Scenario Maker page where users can perform network edits and begin the scenario generation process.

### 3.1.2 Spatial and Behavioural Assessment

The requirement to understand the spatial extent of the impacts infrastructure interventions produce required the display of results spatially disaggregated across multiple geographic scales. Specifically, FASTrack reports changes at the Greater Bendigo level and within 3 km and 1 km scenario catchments, while road-link and SA1-level outputs provide additional spatial detail. This allows users to distinguish broader municipal effects from changes concentrated around an intervention and its immediate surrounding area.

FASTrack also retains behavioural and demographic detail within the scenario outputs. Mode share and mode shift results can be examined by demographic group and trip purpose, while network-level outputs provide information on changes in cycling volumes, travel time, and LTS. Together, these outputs allow users to examine not only whether a scenario produces a change in cycling behaviour, but also where those changes occur and how they vary across different population and trip characteristics.

The resulting decision-support outputs provided by FASTrack are summarised in Table 3. Rather than requiring users to interpret static model outputs, the Scenario Viewer presents the results through interactive maps, summary indicators, and spatial visualisations that connect network changes with behavioural and environmental outcomes.

### 3.1.3 Decision-Support and Investment Appraisal

The requirement for business-case evidence resulted in the integration of economic appraisal directly within the scenario assessment workflow. FASTrack provides interactive CBA incorporating NPV, BCR, and various component benefits. An optional land-use sensitivity component allows users to examine potential additional benefits under user-defined assumptions.

The need to communicate and reuse results was addressed through the Scenario Viewer, automated scenario reports, and downloadable spatial and summary outputs offered by the Scenario Report. The dashboards present modelled outcomes in a form intended to support interpretation by planners and decision-makers, while the automatically generated PDF reports provide a retained summary of the scenario suitable for communication and further consideration. Together, these functions extend FASTrack beyond scenario generation and modelling to provide an end-to-end workflow from infrastructure concept development through assessment,

**Table 3:** Summary of FASTrack decision-support outputs available in the Scenario Viewer and their relevance to cycling infrastructure planning and appraisal.

| Decision-support objective                     | FASTrack feature                                                                                                                              | Planning relevance                                                                                                                                                 |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assess network and infrastructure change       | Interactive maps of cycleway changes and Level of Traffic Stress (LTS), together with LTS distribution summaries                              | Shows how proposed infrastructure changes cycling conditions and network quality within and around the intervention area                                           |
| Assess behavioural response                    | Mode share and mode shift outputs across Greater Bendigo and scenario catchments, with breakdowns by demographic group and trip purpose       | Shows potential changes in travel behaviour and how responses vary across population groups, trip purposes and spatial scales                                      |
| Understand the spatial distribution of impacts | Maps of changes in cycling volume and travel time, together with spatial distributions of trip origins and destinations                       | Identifies where intervention effects are concentrated and how they vary across the study area                                                                     |
| Support economic appraisal                     | Interactive cost-benefit analysis including NPV, BCR and component benefits, with optional land-use sensitivity tools                         | Translates modelled transport impacts into appraisal measures relevant to early-stage business cases, funding applications and investment decisions                |
| Characterise environmental outputs             | Spatial outputs for selected transport-related emissions, including CO <sub>2</sub> , PM <sub>2.5</sub> , NO <sub>2</sub> and NO <sub>x</sub> | Provides complementary information on environmental conditions associated with infrastructure scenarios                                                            |
| Communicate, share and reuse results           | Interactive dashboards along with downloadable spatial datasets, summary outputs, and formatted PDF reports                                   | Supports interpretation and communication of results to decision-makers and stakeholders, while allowing outputs to be retained for reporting and further analysis |

appraisal, and communication.

### 3.2 Bendigo Low Line Demonstration

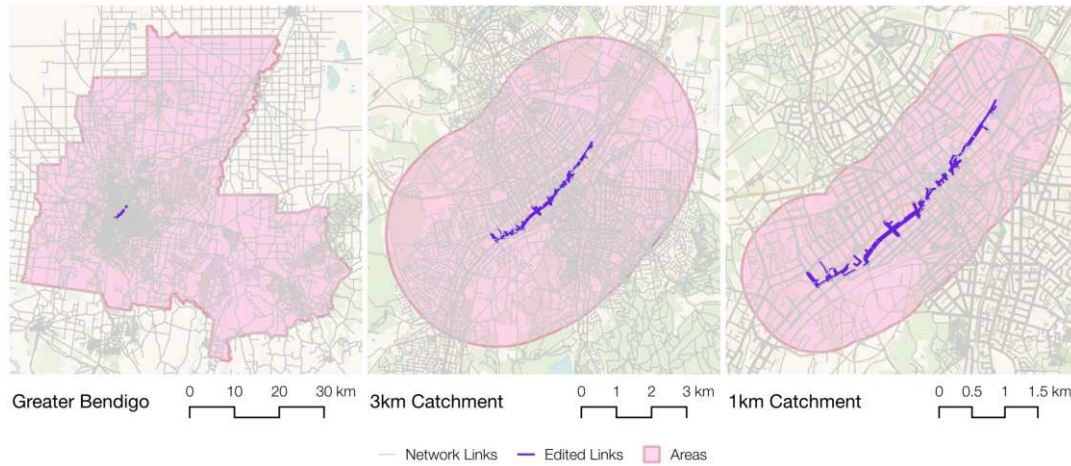
The Bendigo Low Line is an ongoing infrastructure project that will provide a dedicated active transport corridor, enabling safe and efficient off-road travel into and out of the Bendigo city centre. This section uses the Bendigo Low Line as a scenario case study to demonstrate the application of the co-designed FASTrack workflow.

The resulting outputs demonstrate how the intervention can be assessed at different spatial scales, as illustrated in Figure 5. Table 4 shows LTS changes and bicycle mode share across Greater Bendigo and the 3 km and 1 km scenario catchments for only trips with both origin and destination within the corresponding catchment. The magnitude of the estimated changes decreases as the spatial extent of aggregation increases, with the largest changes occurring within the 1 km catchment.

The scenario also demonstrates the integration of behavioural and economic assessment within the same workflow. Only considering trips interacting with the modified network, the probabilistic sum of bicycle mode share increased from 1,164.179 to 1,173.011, an increase of

0.759%, while the corresponding car probability sum decreased from 87,556.814 to 87,550.709, a decrease of 0.007%. Under CBA assumptions specified by local government practitioners for the Bendigo Low Line, the resulting cost-benefit analysis produced a NPV of \$12.53 million and a BCR of 2.56 (Table 5). These results illustrate how the behavioural outputs generated by the modelling workflow can be carried through to economic appraisal without requiring a separate modelling process.

The Scenario Viewer brings these outputs together through a series of dashboards and interactive web maps. Figure 6 illustrates examples of SA1-level cycling travel time changes along with CBA calculator and summary results. The broader set of outputs available through the Scenario Viewer and Scenario Report modules is summarised earlier in Table 3. Together, these outputs demonstrate how a single user-defined intervention can be examined from multiple perspectives within an integrated planning support workflow.



**Figure 5:** Three different spatial aggregation areas for the Bendigo Low Line example case study.

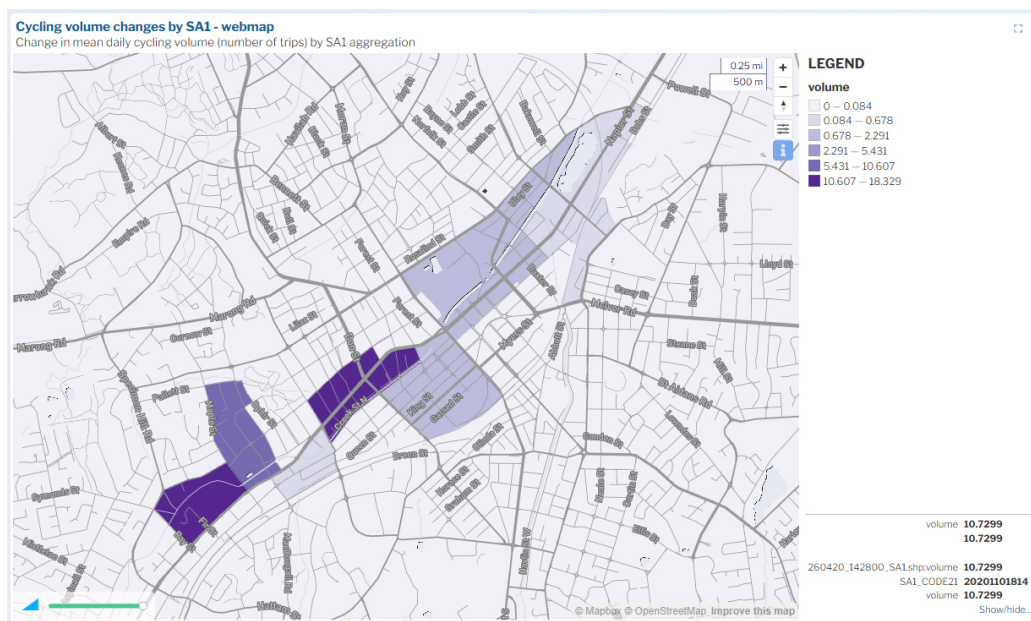
**Table 4:** Changes in impacts of the Bendigo Low Line scenario compared to baseline for all trips with both origin and destination within the corresponding spatial aggregation area.

| Indicator               | Greater Bendigo |           |                 | 3 km Catchment |           |                 | 1 km Catchment |          |                 |
|-------------------------|-----------------|-----------|-----------------|----------------|-----------|-----------------|----------------|----------|-----------------|
|                         | Baseline        | Scenario  | Relative change | Baseline       | Scenario  | Relative change | Baseline       | Scenario | Relative change |
| LTS 1 route proportion  | 38.517%         | 39.388%   | +2.260%         | 24.406%        | 26.254%   | +7.571%         | 13.295%        | 18.976%  | +42.729%        |
| Bicycle probability sum | 6,384.425       | 6,393.260 | +0.138%         | 2,853.154      | 2,862.068 | +0.312%         | 523.267        | 528.757  | +1.049%         |

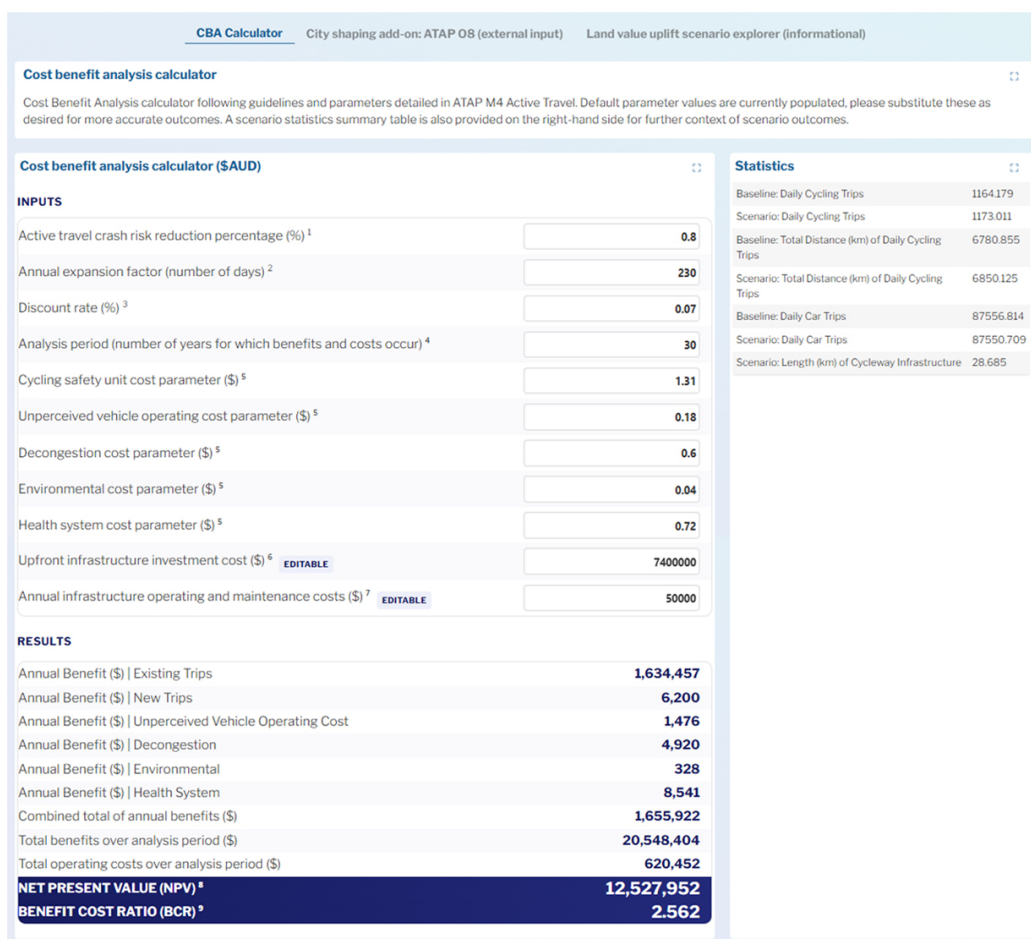
## 4 Discussion

### 4.1 Co-design as a Mechanism for Planning Support

The development of FASTrack demonstrates how co-design can extend beyond conventional user consultation to influence both the functionality and analytical structure of a planning support system. The three-year process combined initial requirements elicitation through user stories with regular practitioner engagement, prototype testing, an extended council trial, and external stakeholder testing. This iterative approach is consistent with research emphasising the value of sustained stakeholder participation in the development of planning support systems, particularly where complex analytical methods need to be translated into practitioner-facing



a.



b.

**Figure 6:** Example outputs from the FASTrack Scenario Viewer for the Bendigo Low Line scenario: a.) changes in cycling travel time spatially aggregated to SA1; and b.) CBA calculator and summary statistics.

**Table 5:** Summary statistics and cost-benefit analysis results for the Bendigo Low Line scenario using an analysis period of 30 years and an active travel crash risk reduction parameter of 0.8 as per ATAP M4.

| CBA indicators                                                 | Value           |
|----------------------------------------------------------------|-----------------|
| Bicycle probability sum (baseline)                             | 1,164.179       |
| Bicycle probability sum (scenario)                             | 1,173.011       |
| Relative change in bicycle probability sum (edited-link users) | +0.759%         |
| Car probability sum (baseline)                                 | 87,556.814      |
| Car probability sum (scenario)                                 | 87,550.709      |
| Relative change in car probability sum (edited-link users)     | −0.007%         |
| Upfront infrastructure investment cost                         | \$7.4 million   |
| Annual operating and maintenance cost                          | \$50,000        |
| Net Present Value (NPV)                                        | \$12.53 million |
| Benefit-Cost Ratio (BCR)                                       | 2.56            |

platforms (Rittenbruch et al., 2022; Blomkamp, 2018; Wacnik et al., 2025). An important characteristic of the FASTrack process was that requirements could be revisited as the system developed, allowing practitioners to respond to working prototypes rather than attempting to define all system requirements at the outset.

This provides a useful distinction from approaches in which stakeholder engagement primarily occurs after the underlying analytical system has already been established. In FASTrack, practitioner input was used to define the decision-support problem, while the research team determined how these requirements could be addressed technically. Co-design therefore acted as an intermediary between practitioner knowledge and technical development, rather than requiring practitioners to directly specify modelling methods. This distinction is particularly relevant to planning support systems based on complex computational models, where the technical expertise required to develop a model may differ substantially from the expertise required to apply its outputs in practice.

## 4.2 Practitioner Priorities and Cycling Planning Systems

The co-design process identified priorities centred on rapid scenario testing, accessible scenario creation, representation of different infrastructure types, spatially detailed assessment, business-case evidence, clear communication of results, and access without specialist software or computing resources. These priorities suggest that rather than discrete modelling components, practitioners require an integrated workflow through which infrastructure concepts can be developed, assessed, interpreted, and communicated.

These findings align with the broader development of cycling planning platforms such as the Propensity to Cycle Tool and CRUSE, which demonstrate a growing shift towards interactive and scenario-based approaches to cycling infrastructure planning (Lovelace et al., 2017; Lovelace et al., 2024). However, this study’s contribution is more focused on how practitioner requirements can shape the design of the underlying planning support system. This is particularly relevant given recent research highlighting the increasing sophistication of cycling modelling alongside continuing challenges in representing the behavioural and environmental factors that influence cycling (Bostanara et al., 2025). As modelling becomes more sophisticated, ensuring that its complexity remains appropriate to the decision-making context becomes

increasingly important.

### 4.3 Translating Practitioner Needs into Design Solutions

The identified priorities were translated into interconnected decisions concerning FASTrack’s modelling approach, software architecture, interface, and decision-support outputs. The most consequential example was the requirement for rapid scenario assessment, which informed the adoption of a probabilistic mode-choice and route-choice approach rather than repeatedly executing the full iterative agent-based model for every scenario. This represents a deliberate trade-off between behavioural and computational complexity, whereby the probabilistic approach provides relatively rapid scenario assessment whilst still retaining a behavioural basis for estimating changes in cycling and other transport choices.

Importantly, co-design did not require practitioners to determine the modelling methodology itself. Rather, practitioners identified the decision-support requirements, while the research team determined how those requirements could be addressed within the technical and methodological constraints of the existing modelling framework. This distinction illustrates how co-design can operate effectively in technically complex planning research: practitioner knowledge defines the problem and desired use of the system, while technical expertise translates those requirements into appropriate analytical and computational solutions.

The same principle extended across the broader system design. The requirement for accessible scenario creation informed the interactive Scenario Maker; the need to understand spatial consequences resulted in multi-scale network and area-based outputs; business-case requirements led to the CBA and land-use benefits integration; and the need for interpretable and communicable results informed the interactive dashboards and automated reporting. The significance of these decisions is therefore not the individual features themselves, but the way they collectively translate practitioner requirements into a coherent decision-support workflow. FASTrack demonstrates that co-design can influence not only what a planning support system presents to users, but also the level of analytical complexity and computational infrastructure required to produce those outputs.

### 4.4 Implications for Regional Cycling Planning

The focus on a regional city is an important aspect of FASTrack’s contribution. Existing cycling planning support systems have often been developed around metropolitan contexts where extensive datasets, modelling infrastructure, and specialist analytical capacity are more readily available. FASTrack instead demonstrates how a comparatively complex behavioural modelling framework can be packaged into an accessible scenario assessment workflow for a regional planning context. The co-design process was important in this translation because focused attention on what practitioners needed to assess and communicate when developing cycling infrastructure proposals.

For regional governments with relatively limited modelling capacity compared to metropolitan contexts, this type of planning support system may provide a means of accessing behavioural and economic evidence without requiring dedicated modelling infrastructure or specialist technical expertise. However, FASTrack should be viewed as a complement to detailed transport modelling and project appraisal, rather than an entire replacement. The primary value of FASTrack is seen in the enabling of rapid exploration and comparison of infrastructure scenarios during early stages of planning and investment development.

### 4.5 Limitations and Future Research

Several limitations should be considered when interpreting the results. First, the co-design process was conducted primarily with government practitioners involved in the project, meaning

that the priorities identified may not represent the full range of requirements across regional governments, community organisations, or other cycling stakeholders. Although the later practitioner trial and Bicycle Network workshop provided broader perspectives, further application across different regional contexts is required to assess the transferability of both the identified priorities and the co-design approach. Future research could examine the development and application of FASTrack across additional regional cities and infrastructure contexts, as well as evaluate how practitioners use co-designed planning support systems in actual planning and investment decisions. Such applications would provide further evidence regarding which aspects of the co-design process are context-specific and which may provide more generalisable principles for developing practitioner-facing planning support systems.

Second, the FASTrack modelling workflow provides a rapid probabilistic assessment of behavioural responses rather than executing a full multi-iteration agent-based model for each scenario. This computational efficiency is central to the FASTrack’s intended use, but it also represents a trade-off between processing speed and modelling complexity. The outputs should therefore be interpreted as scenario assessment estimates rather than forecasts of realised travel behaviour. Future work however could seek to optimise the computational efficiencies of cloud-based multi-iteration agent-based modelling for more rapid processing and through scenario modelling outputs.

Finally, limitations relate to the source data and behavioural assumptions underlying the B-ABM. The cycling behavioural components were primarily informed by commuting and journey-to-work<sup>2</sup> trips from the VISTA 2012–2023 travel surveys (Victorian State Government Department of Transport and Planning, 2024) and data collected through the cycling route-choice public participatory map-based survey (Tiwari et al., 2025), because sufficient observations were not available for other cycling trip purposes. This limitation also informs the default parameters of the CBA calculator, which uses a calendar year of 230 working days to represent journey-to-work trips and trips undertaken for employer business. Consequently, FASTrack scenario outputs do not comprehensively represent non-work-related cycling purposes, such as recreational trips, which may also respond to cycling infrastructure interventions (Bostanara et al., 2025). Expanding the underlying behavioural data to capture a broader range of cycling purposes could therefore be an important direction for future development.

## 5 Conclusion

This study demonstrates how co-design can translate practitioner requirements into the methodological, technical, and functional design of a complex planning support system. Through a three-year iterative process of requirements elicitation, practitioner engagement, prototype testing, and stakeholder evaluation, FASTrack was designed to provide rapid, accessible, spatially informative, and investment-relevant cycling infrastructure assessment. These requirements shaped the development of its probabilistic modelling approach, cloud-based workflow, interactive scenario development, spatial outputs, cost-benefit analysis, and reporting functions. The broader contribution is therefore not the individual modelling or software components, but the demonstration of how sustained co-design can bridge practitioner needs and technical system development, providing a transferable approach for ensuring that increasingly sophisticated planning support systems remain aligned with the practical requirements of decision-makers.

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<sup>2</sup>Commute and journey-to-work data is accessible from <https://www.abs.gov.au/articles/australias-journey-work>

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## 7 Declaration of Conflicting Interest

The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## 8 CRediT author contribution statement

**Ryan Turner:** Conceptualization; Methodology; Software; Formal analysis; Data Curation; Writing - Original Draft; Writing - Review & Editing; Visualization. **Afshin Jafari:** Conceptualization; Methodology; Software; Formal analysis; Data Curation; Writing - Original Draft; Writing - Review & Editing; Supervision; Project administration; Funding acquisition. **Sapan Tiwari:** Conceptualization; Methodology; Software; Formal analysis; Data Curation; Writing - Original Draft; Writing - Review & Editing. **Alan Both:** Conceptualization; Methodology; Writing - Review & Editing; Supervision. **Dhirendra Singh:** Conceptualization; Methodology; Writing - Review & Editing; Supervision. **Steve Pemberton:** Conceptualization; Methodology; Writing - Review & Editing. **Chris De Gruyter:** Conceptualization; Methodology; Writing - Review & Editing.

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## 10 Data and code availability

Code repositories of open-access modelling components used in this study are provided as follows: Network generation <https://github.com/rmit-astm/network> Cycling safety and comfort <https://github.com/rmit-astm/cycling-safety-comfort>

## 11 Declaration of Generative AI

The authors declare that OpenAI ChatGPT-5.6 was used solely as a language-editing aid to improve the readability, grammar, and stylistic clarity of the manuscript. Generative AI was not used to develop or alter the scientific ideas, conduct or modify the data analysis, interpret the results, or formulate the conclusions presented in this study.

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